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Spiking Reservoir Computing Architectures for Model-based Epileptic Brain State Recognition

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***Abstract*— Bio-inspired networks offer rich dynamic capabilities with minimal energy demands, especially when implemented on neuromorphic hardware. In particular, the recurrence in brain circuits enables Recurrent Spiking Neural Networks (RSNNs) to generate complex spatio-temporal spike patterns, forming internal representations of time-varying signals. Despite this biological sophistication, such architectures are often applied to machine learning tasks with limited biological relevance. In this work, we present a spiking reservoir computing architecture, implemented as a fully spiking Liquid State Machine (LSM) with Leaky Integrate-and-Fire (LIF) neurons, designed to recognize and decode internal brain states. We simulate epileptic activity using a spiking model and develop a complete pipeline where a source generates Local Field Potentials (LFPs). These signals are encoded through population coding and processed by the LSM, which performs regression on the biophysical parameters controlling epileptic dynamics, thereby inferring the source state. To enhance the LSM's performance, we apply a biologically inspired synaptic plasticity mechanism to the RSNN. Our results demonstrate that a simple, unsupervised plasticity mechanism can optimize the internal parameters of the reservoir, particularly in smaller networks. This approach offers a hardware-efficient strategy for task-specific adaptation of general-purpose circuits, highlighting its suitability for edge-device implementations. Our findings emphasize the potential of spiking reservoir computing for real-time decoding of complex brain dynamics, such as epileptic activity, and underscore the advantages of biologically inspired, energy-efficient methods for neuromorphic systems in real-world applications.**

***Clinical relevance*— The ability to decode internal brain states from Local Field Potentials (LFPs) using a biologically inspired spiking reservoir computing architecture has significant implications for clinical neuroscience. By providing a real-time, energy-efficient method for tracking epileptic dynamics, this approach could aid in the development of advanced brain-computer interfaces (BCIs) and personalized neurostimulation therapies. Such a system may help clinicians monitor seizure progression, optimize treatment strategies, and improve patient outcomes in clinical epilepsy management.**

I. INTRODUCTION

Reservoir Computing (RC) is a computational paradigm inspired by recurrent neural networks (RNNs) that has gained attention for its ability to process time-varying inputs efficiently [1], [2], [3]. It leverages the computational power of a randomly initialized, fixed recurrent neural network, called a reservoir, to project input signals into a high-dimensional space where they can be easily analyzed and processed by a simple linear readout layer. This approach simplifies the training process and reduces computational costs compared to traditional RNNs, making it attractive for various applications, including time series prediction, signal processing, and robotics [4], [5], [6].

One prominent RC model is the Liquid State Machine (LSM), figure 1, which is biologically inspired and aims to mimic the information processing capabilities of cortical microcircuits in the brain. LSMs consist of a recurrent network of spiking neurons, known as the liquid layer or *reservoir*, where the connections between neurons are typically fixed and randomly initialized. The liquid's response to input signals is then read out by a linear classifier or regressor to perform specific tasks.

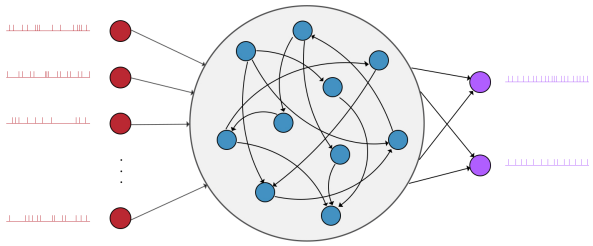

Fig. 1. Liquid State Machine architecture. Input neurons (red circles, left) project signal into the reservoir (grey circle, middle), that is a recurrent network. Readout neurons (purple circles, right) receive reservoir signal and are trained to produce a spiking pattern that solves the task.

A fundamental property of LSMs, crucial for their proper functioning, is the separation property [2]. This property ensures that the liquid can generate distinct internal states in response to different input signals. In other words, it guarantees that the reservoir can effectively discriminate between various inputs, enabling the readout layer to learn and perform accurate computations.

In this study, we explore the potential for enhancing LSM performance by incorporating Spike-Timing-Dependent Plasticity (STDP), a biologically plausible learning rule observed in the brain. STDP adjusts the strength of synaptic connections between neurons based on the precise timing of their spikes. By introducing STDP into the recurrent connections of the reservoir, we aim to optimize the network's dynamics

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and improve its computational capabilities.

Specifically, we investigate how STDP on the separation property of the LSM. We hypothesize that by allowing the recurrent connections to adapt through STDP during an unsupervised exposure phase, we can fine-tune the reservoir's internal representations and enhance its ability to separate different input patterns. This, in turn, could lead to improved performance in downstream tasks.

To validate our hypothesis, we apply the STDP-optimized LSM to a biologically relevant task: the recognition of epileptic seizures. Epilepsy is a neurological disorder characterized by recurrent seizures, which are caused by abnormal, synchronized electrical activity in the brain. We utilize a synthetic model of epileptic neural activity, known as the Epileptor [7], [8], to generate simulated brain signals. The task of the LSM is to predict and control the dynamics of the Epileptor based on these signals.

By comparing the performance of the STDP-optimized LSM with that of a standard LSM with fixed random connections, we aim to demonstrate the potential benefits of incorporating biologically inspired plasticity into reservoir computing. We anticipate that the STDP-optimized LSM will exhibit superior performance in controlling epileptic seizures due to its refined internal representations.

In the following sections, we will detail our methodology, including the implementation of STDP, the experimental setup for the seizure control task, and the evaluation metrics used to assess performance. We will then present our results, highlighting the impact of STDP on the separation property and the overall performance of the LSM in controlling epileptic seizures. Finally, we will discuss the implications of our findings and potential future directions for research in this area.

II. METHODS

The computational power of Liquid State Machines (LSMs) is determined by the properties of the liquid layer, which strongly depend on the set of recurrent connection weights that remain fixed during the system's training. In this study, we apply STDP plasticity rule to the recurrent synapses, recognizing that the resulting weight changes may affect the necessary properties of the liquid layer.

A. Neuronal model

Our Spiking Neural Network employs Leaky Integrate-and-Fire (LIF) neuron models, widely recognized for its balance between biological plausibility and computational efficiency. It captures essential neuronal dynamics, such as membrane potential decay and spike generation, but remains computationally tractable for large simulations or neuromorphic hardware. This simplicity facilitates efficient simulations in both software and hardware implementations.

More specifically, we utilize the Current-Based (CuBa) version of the LIF model with fixed threshold and decaying-exponential post-synaptic current, defined as in 1, equation which describes the subthreshold dynamics of the membrane potential:

$$\dot{V}(t) = -\frac{V(t) - V_{rest}}{\tau_m} + \frac{I_{syn}(t)}{C_m} \tag{1}$$

Here, $V(t)$ represents the membrane potential at time t, while V_{rest} is the resting membrane potential, which is the baseline voltage of the neuron in the absence of synaptic inputs. The parameter τ_m is the membrane time constant, C_m is the membrane capacitance. The term $I_{syn}(t)$ represents the synaptic input current at time t, which modulates the membrane potential based on external stimuli.

The synaptic current I_{syn} injected into the $j-$th neuron is given by:

$$I^j_{syn}(t) = \sum_i J_{ij} S_i(t)\ e^{-t/\tau_{syn}} \tag{2}$$

Here, j is the index of the postsynaptic neuron, i is the index of presynaptic neurons, J_{ij} is the synaptic weight, S_i is the firing state of the presynaptic neuron i and τ_{syn} is the decay time of the synapse.

Parameters for the LIF models are included in the table III. Neuron and synapse models have been simulated using pyNN [9], with NEURON [10] as the background simulator.

B. Initialization of the liquid layer

To make sure that our networks start with the correct properties in terms of stability, we initialize the connectivity matrices (i.e., the initial synaptic weights between neurons in the reservoir) following the rule by [11], that is universal and controls the spectral content of the matrices.
The steps for the initialization are:

1) Start with a random matrix W with all non-negative entries
2) Scale W so that $\rho(W) < 1$
3) Change the signs of a desired number of entries of W to get negative connection weights as well

Here, $\rho(W)$ indicates the spectral radius if the matrix W. This recipe ensures that the defined reservoir is stable with respect to *any* input signal, thus serving as a general purpose, universal (and possibly strict) initialization.

The initial entries of the matrix are randomly sampled from a normal distribution with zero mean, while the variance of the distribution scales with the size N of the reservoir as $\sigma^2 \sim 1/N$, while the weights are expressed in units of current in order to stimulate the CuBa LIFs (see sec. II-A).

The sign of each entry reflects the nature (excitatory/inhibitory) of the pre-synaptic neuron from which the corresponding synapse originates. In our paper, we respect Dale's principle [12][13]: inspired by biological neural networks, this principle dictates that a neuron releases the same set of neurotransmitters at all of its post-synaptic connections. In our implementation, we enforce Dale's principle by ensuring that each neuron in the reservoir is either excitatory or inhibitory, and its connections to other neurons are assigned weights accordingly.

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C. Separation Property

The Separation Property (SP) is is a crucial requirement for the proper functioning of LSMs. SP refers to the ability of the liquid layer to produce significantly different internal states when stimulated by significantly different inputs.

This property depends on the specific input provided. In this work, we use SP tests to assess the stability of the reservoir when stimulated with the epileptiform LFP input signal described in II-F.

We evaluate the SP by considering pairs of input stimuli u and v, given in the form of spike trains with distances $d(u, v)$ between them and examining whether the corresponding liquid states, X_u and X_v, exhibit a similar trend in their respective distances $d'(X_u, X_v)$.

When this property is respected, we know that the reservoir is in the correct stable regime and can work as a good source for the readout that, in turn, can be trained to perform different tasks.

In our experiment, we defined d as in the original paper [2]: a (squared) L_2 norm between the two gaussian-filtered spiketrains:

$$d(u, v) = \frac{1}{T} \sum_{t=1}^{T} (\tilde{u}_t - \tilde{v}_t)^2 \Delta t \tag{3}$$

Here, $\tilde{u}_t$ represents the spiketrain convolved with a gaussian kernel $e^{-(t/\tau)^2}$ at time t. The chosen kernel has $\tau = 5\ ms$, while T is the total duration of the stimulus ($0.5\ s$) and $\Delta t = 0.1\ ms$ is the time resolution in the spiketrains.

For the distance d' between the liquid states, we used the Euclidean Distance defined as follow:

$$d'(X_u, X_v) = \frac{1}{N} \sum_{n=1}^{N} (\hat{X}_u^n - \hat{X}_v^n)^2 \tag{4}$$

Here, $\hat{X}_u^n$ represents the exponentially filtered spiketrain of the n-th neuron in the reservoir. The exponential kernel is defined as $e^{-(t/\tau)}$ and the chosen time constant is $\tau = 5\ ms$.

To check if a given reservoir shows the SP, we check that the patterns of $d'(X_u, X_v)^P$ in the plastic state are statistically similar to the ones in the initial stable state $d'(X_u, X_v)^0$:

$$\frac{\langle d'(X_u, X_v)^P \rangle_N}{\langle d'(X_u, X_v)^0 \rangle_N} \simeq 1 \tag{5}$$

where $\langle \cdot \rangle_N$ indicates the average over the N reservoir neurons.

D. Plasticity protocol

In the reservoir computing framework, the recurrent weights are kept constant after initialization and during the task. This has the advantage of speeding up the training by avoiding the need to update the largest portion of the model's synapses, other than allowing the model for multitasking: by limiting the training only at the readout phase, the model is able to keep generality at the reservoir phase, effectively working as an unbiased internal representation of the input signal that can be read out in different ways according to different needs.

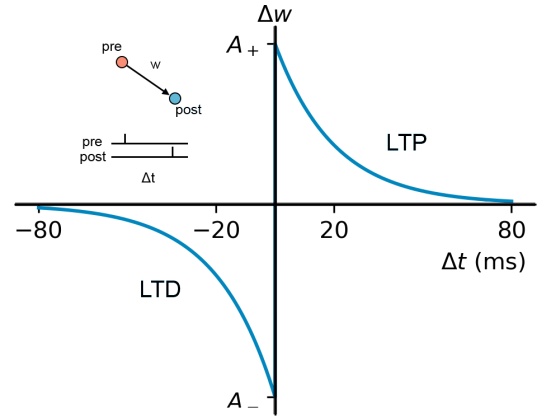

Fig. 2. Spike Timing Dependent Plasticity window. Short delays in the post-synaptic neuron firing leads to increases (Long Term Potentiation, LTP) in the weight w connecting the corresponding pre-post pair. Small anticipations in the post-firing leads to weight depression (Long Term Depression, LTD).

At the same time, this generality represents a limitation for each specific task: given the unbiased nature of the representations that the reservoir generates, each specific task is performed in an under-optimized manner by using unspecific representations. Extending training also at the reservoir level can improve task performance and is the standard approach in Artificial Neural Networks, but this typically leads to biologically unrealistic plasticity rules that depend strongly on the given task.

Here, we demonstrate that an intermediate approach that can be used in the pre-training phase, working as an effective, unsupervised and input-dependent optimization of the reservoir synapses. Being unsupervised, it maintains the generality with respect to any input signal and specific task.

Although different approaches are used in the literature to achieve such optimization [14], [15], [16], [17], [18], [19], we propose that simple STDP plasticity works when small networks and fast elaborations are involved, representing the prototypical candidate for such plasticity-based optimization protocols without adding extra mechanisms that are usually involved to make more realistic but case-specific models of plasticity.

The STDP rule is the one of the eq. 6 and Fig. 2, where a potentiation in the synaptic strength can be observed for positive delays in the firing of the post-synaptic neuron, while a depression is measured for negative values of the pre-post firing times Δt.

$$\Delta J_{ij} = \begin{cases} A_+ e^{-\Delta t/\tau_+} & \Delta t > 0 \\ A_- e^{-\Delta t/\tau_-} & \Delta t < 0 \end{cases} \tag{6}$$

We apply STDP for a given amount of time that we call exposure time. We set different values of the exposure time for each network size and study its impact on SP.

The chosen parameters of the STDP are included in the Tab. III.

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E. Coding

To interface the analog LFP signals with the spiking neural network of the LSM, we utilize a population coding protocol. Population coding is a method for representing analog values using the firing patterns of a population of neurons, setting a linear relation between the analog value and the position of the coding neuron. In our implementation, similar to the one present in [20], we divide the range of possible normalized LFP values into discrete intervals and assign each interval to a specific neuron in the input layer of the LSM. The firing rate of each input neuron is then proportional to the probability that the LFP signal falls within its assigned interval.

By employing population coding, we effectively convert the continuous LFP signals into a format that can be processed by the spiking neurons in the LSM. This allows us to leverage the computational power of the LSM to analyze and interpret the complex dynamics of epileptic seizures.

At the same time, the output of the readout neurons need to be converted in order to be interpreted as values for the epileptor parameters. To do this, we employed a Firing Rate coding step, in which the frequency of readout firing is linearly mapped to a parameter value in a defined range.

F. Application to biologically relevant task

When the SP is respected, we can train the resulting LSMs to perform different tasks. In this work we focus on epileptiform activity identification in seizure-expressing populations of neurons. This will allow us to evaluate the performance of the STDP-optimized LSM in a real-world scenario, while applying it to a biologically relevant task: the control of epileptic seizures.

We utilize the Epileptor model [7], a well-established computational model of epileptic neural activity, to simulate seizures. The Epileptor model captures the dynamics of seizure generation and propagation in the brain, providing a realistic testbed for our approach.

More specifically, we implement the computational model of the Epileptor based on Spiking Neural Networks described in [8]. From this model, we extract synthetic Local Field Potential (LFP) signals, which represent the electrical activity at the small scale of the brain during a seizure. These signals serve as the input to the LSM, and the task is to predict the dynamics of the Epileptor based on these LFP signals. Specifically, the LSM aims to estimate the values of two biophysical parameters that control the behavior of the Epileptor and, in turn, the physiopathological state of the network.

To achieve this, we employ a regression task where the LSM learns to map the LFP signals to the corresponding parameter values. We train the LSM using supervised learning, where the ground truth parameter values are provided during the training phase. The performance of the LSM is then evaluated based on its ability to accurately predict the parameter values on unseen LFP signals.

The LFP signal is divided into slices and each training step is performed slice by slice, while the learning rule for the task is reported in 7 and it gives the updates for the reservoir to output weights J^{out} for each example presentation:

$$\Delta J_{ij}^{out} = \eta(\mathbf{p}_j^* - \mathbf{p}_j)\hat{\mathbf{S}}_i^T \tag{7}$$

Where i is the index of the presynaptic reservoir neuron ($i = 1, ..., N$) and j is the readout neuron index (one for each parameter, $j = 1, 2$). The weight update rule 7 is an application of the delta rule [21], [2], where $\mathbf{p}^* = (x^*, y^*)$ represents the target couple of parameters, while $\mathbf{p} = (x, y)$ is the actual value of the parameters. Both x and y are found by linearly mapping the readout firing rates into parameter values. The difference between actual and target parameters represents the error term, that is multiplied by the (exponentially filtered) synaptic input $\hat{S}_{res}(t)$ coming from the reservoir and η is a multiplicative factor.

Considering that firing rate estimates are made for each slice of LFP signal injected in the network (in our case after collecting 500 ms of data, see Table III), the rule 7 collectively considers the sum over time of each reservoir spiketrain $\hat{\mathbf{S}}_{res}^T$, effectively providing a single update value (with sign) for each readout weight that does not explicitly depend on time.

Training and task parameters are included in the table III, while a schematic wiew of the experiment is included in the figure 3.

III. RESULTS

In this section we show the results of the regression task in both plasticity enabled and standard scenarios. The task aims to produce proper spiking patterns in the readout neurons that, once decoded, represent the values of the couple of parameters.

To evaluate performance in solving the task, a custom loss 8 has been defined that measures the euclidean distance in the parameter space between the predicted and true coordinates.

$$\mathcal{L} = \sqrt{(x^* - x)^2 + (y^* - y)^2} \tag{8}$$

Here, (x, y) are the predicted coordinates decoded from the firing rate of readout neurons and (x^*, y^*) are the ground truth coordinates.

It is important to notice that we use 8 only to assess task performance but this is not directly the target function to be minimized. The training objective is to produce proper firing rates that minimize the difference between target and actual spiketrains, and this, in turn, lead to the minimization of the distance between parameter coordinates.

The results for the regression task are reported in the table I and in figure 4. The results are shown in the two reservoir cases: standard LSM and STDP-enabled with the best performing exposure time.

For each reservoir size, we observe that an optimal plasticity exposure time is present and its value depends on network size. For relatively large networks, even small exposure times can lead to cases in which SP is not respected, and STDP optimization needs to be carefully tuned. In the small-network case, STDP is beneficial in the majority of cases.

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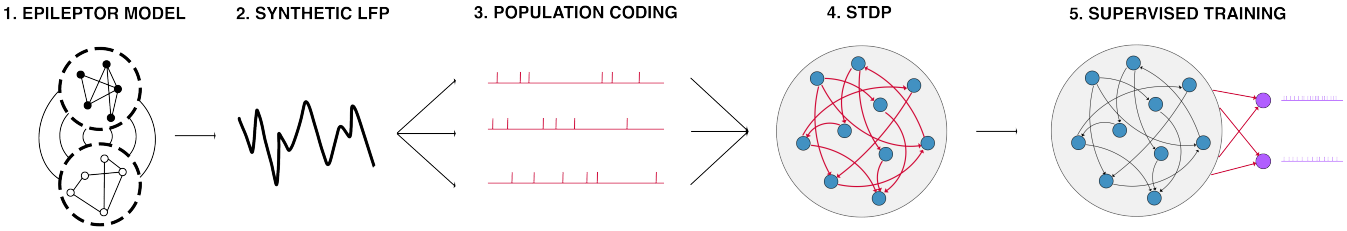

Fig. 3. Experimental setup. The Epileptor Model (**1**) simulates population of neurons that produces epileptiform activity. Signal from the Epileptor neurons is collected in the form of a Local Field Potential (LFP, **2**) and then passed through a population coding coding process (**3**), that results in spiking patterns of input neurons. The obtained spiking signal is then projected into the STDP-enabled liquid layer (**4**) that, after an *exposure time* in which recurrent weights are free to change according to the plasticity rule, is finally frozen and then trained in a supervised way to perform regression on epileptor parameters (**5**).

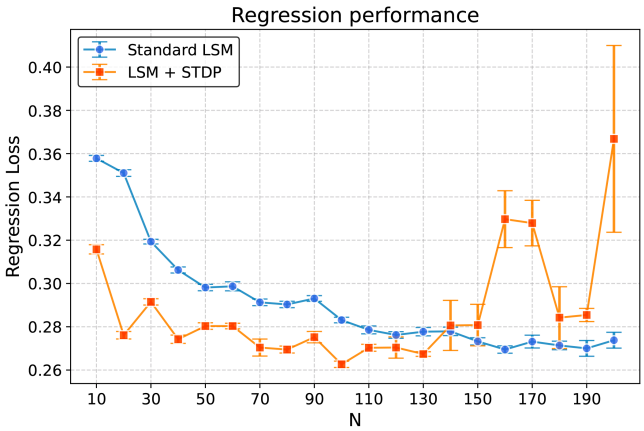

Fig. 4. Regression performance for both standard LSM (blue) and STDP-enabled LSM (orange). Each point represents the average loss of the equation 8 of ten randomly initialized reservoirs (ten seeds). Numeric values are reported in the table I.

TABLE I

NETWORK PERFORMANCE ON THE REGRESSION TASK

N	**10**	**20**	**30**	**40**	**50**
LSM	0.358	0.351	0.319	0.306	0.298
LSM+STDP	**0.316**	**0.276**	**0.291**	**0.274**	**0.280**
N	**60**	**70**	**80**	**90**	**100**
LSM	0.291	0.290	0.293	0.283	0.279
LSM+STDP	**0.280**	**0.270**	**0.269**	**0.275**	**0.262**
N	**110**	**120**	**130**	**140**	**150**
LSM	0.279	0.276	0.278	**0.278**	**0.273**
LSM+STDP	**0.270**	**0.270**	**0.267**	0.281	0.281
N	**160**	**170**	**180**	**190**	**200**
LSM	**0.269**	**0.273**	**0.271**	**0.270**	**0.274**
LSM+STDP	0.330	0.328	0.284	0.285	0.367

The relative impact on network performance due to STDP for all combinations of network size and exposure times can be found in the figure 5, where the heatmap shows the percentage decrease (blue) or increase (red) of the loss when STDP is applied for the corresponding time with respect to the regular, non-plastic reservoir. All results have been averaged over ten different reservoir initializations.

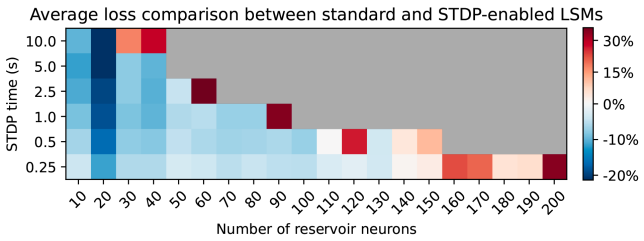

Fig. 5. Performance comparison between plasticity-enabled reservoirs and standard ones. The color codes for the increase (red) and decrease (blue) in percentage of the regression loss 8 when using STDP-enabled reservoirs, with negative values indicating an increase in performance (the lower, the better). Grey area corresponds to combinations of parameters that produce ill-defined reservoirs that don't respect Separation Property 5 and no results are shown in these cases.

Given the biological interpretation of the parameters found by the regression, we can assess network capability in identifying the epileptic state of the data. The parameter space can be partitioned into four quadrants, each indicating a state of developing epileptic seizure and the characterization of the Status Epilepticus (SE):

- Zone I - before SE (physiological regime).
- Zone II - impending SE.
- Zone III - established SE.
- Zone IV - subtle SE.

Once defined the four zones, we can measure network performance in a derived classification task in which each zone represents a class. We divide the space of explored parameters with the mean value of each coordinate obtaining the four zones, and the classification metrics are reported in table II for the best performing networks in the plastic/standard case (the ones with performance underlined in table I).

We can see that, while the full physiological and epileptic zones can have similar performance in the two cases, zones II and IV exhibit notable gains in F1-score with the STDP-enabled LSM, mainly driven by improved recall. Interestingly, this improvement is observed despite a smaller network size (100 neurons versus 160 in the best-performing standard LSM, see table I). Enhanced performance in zone II can aid in anticipating seizure onset and potentially intervening in the tissue to restore physiological activity. Bootstrap analysis of classification accuracy further indicates

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TABLE II

NETWORK PERFORMANCE ON THE CLASSIFICATION TASK.

Zone	Model	Precision (%)	Recall (%)	F1-score (%)
I	LSM	80.6	**92.2**	86.0
	LSM+STDP	**93.0**	88.9	**90.9**
II	LSM	68.4	68.1	68.3
	LSM+STDP	**68.6**	**77.1**	**72.7**
III	LSM	73.3	**86.2**	79.2
	LSM+STDP	**77.1**	84.8	**80.7**
IV	LSM	**100.0**	45.6	62.6
	LSM+STDP	**100.0**	**52.2**	**68.6**

a performance trend in favor of the STDP-enabled model, with a mean accuracy of 77.8% (±1.6%, 95% Confidence Interval (CI): [74.7–80.8]%), compared to 74.7% (±1.8%, 95% CI: [71.2–78.2]%) for the baseline LSM. Although the confidence intervals slightly overlap, the observed difference suggests a benefit of incorporating synaptic plasticity in spiking models that allows for the use of smaller network sizes.

An example of the output decoded coordinate in the parameter space is provided in figure 6, where each point has the color of the ground truth zone.

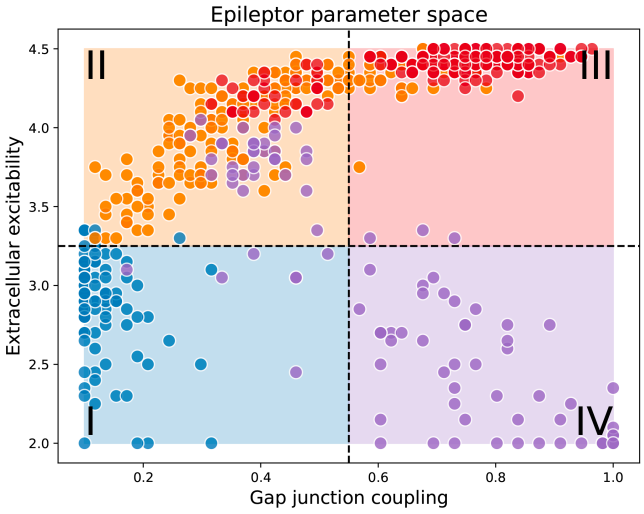

Fig. 6. Classification setting. The parameter space is divided into four quadrants: **I** (blue) - physiological activity; **II** (orange) - impending SE; **III** (red) - established SE; **IV** (purple) - subtle SE. Each point represents the output of one selected model when exposed to the test data (600 points) and the color of each point represents the ground truth zone. Axis values are given for indicative purposes and do not portray biophysical units, as in the original paper [8].

IV. CONCLUSIONS

In this work, we introduced a bio-inspired, energy-efficient, and model-based framework for feature extraction and classification of epileptiform dynamics. By leveraging spiking reservoir computing and a simple STDP rule, we achieved effective representations of Local Field Potential (LFP) signals while preserving clinical interpretability through a model-based mapping onto physiological parameters. Such an approach holds particular promise for neuromorphic implementations, where event-driven updates and parallel architectures can efficiently handle the local nature of STDP.

Despite these advantages, relying solely on a purely Hebbian plasticity rule can create instabilities within the reservoir. Without additional mechanisms (e.g., homeostatic controls, metaplasticity, or constraint-based weight scaling), runaway potentiation or depression may lead to lost generality or oscillatory states. Nonetheless, from a hardware perspective, simpler STDP mechanisms are easier to implement in plasticity-enabled neuromorphic platforms—such as BrainScaleS-2 [22]—offering a practical balance between biological plausibility and engineering feasibility.

Finally, we note that the concept of "classification" in this study hinges on how one partitions the parameter space of the Epileptor model. Because epilepsy is not strictly bounded by discrete zones, dividing the space into quadrants is merely a convenient heuristic to illustrate the potential for biologically interpretable feature extraction. In reality, any meaningful subdivision should account for both model parameters and real physiological indicators, and future work may refine or replace the quadrant-based approach to align more closely with clinical definitions of seizure states.

Our findings suggest that incorporating a brief, unsupervised STDP phase can significantly reshape the internal dynamics of a Liquid State Machine and enhance its capacity to detect changes in epileptiform signals. By adjusting synaptic strengths in response to spike-timing correlations, the reservoir becomes better aligned to the structure of its input distribution. This biologically plausible adaptation can be especially advantageous in contexts where the input signals share common statistical features (e.g., brain activity during epileptic seizures).

Interestingly, our results show that smaller reservoirs tend to benefit the most from STDP-driven pre-training, with larger improvements in prediction accuracy compared to standard, randomly initialized networks. One potential explanation is that, in small networks, the random initialization may provide insufficient coverage of the input space, leaving more room for STDP to shape synaptic weights toward useful spatiotemporal representations. In contrast, larger reservoirs may already capture diverse dynamics via random connectivity alone, thereby reducing the relative gain from STDP.

Although STDP provides a valuable mechanism for tuning reservoir dynamics, it can also destabilize network activity if applied too aggressively or for an overly long exposure period. Excessive potentiation can lead to runaway effects, causing the reservoir to lose its generality or even violate the separation property. Additionally, task-specific tuning might inadvertently reduce the reservoir's capacity to handle different input domains, thus limiting the general-purpose appeal of reservoir computing. Incorporating homeostatic controls or metaplasticity rules could mitigate these risks in future work.

From a hardware standpoint, the local nature of STDP

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TABLE III

SIMULATION PARAMETERS

Name	Value	Units	Description
Reservoir Neuron Parameters			
V_{rest}	-65.0	mV	Resting membrane potential
C_m	1.0	nF	Capacity of the membrane
τ_m	8.0	ms	Membrane time constant
$\tau_{syn,E}$	2.0	ms	Decay time of excitatory synaptic current
$\tau_{syn,I}$	2.0	ms	Decay time of inhibitory synaptic current
V_{thresh}	-50.0	mV	Spike threshold
V_{reset}	-65.0	mV	Reset potential after a spike
τ_{ref}	0	ms	Duration of refractory period
Readout Neuron Parameters			
V_{rest}	-65.0	mV	Resting membrane potential
C_m	1.0	nF	Capacity of the membrane
τ_m	5.0	ms	Membrane time constant
$\tau_{syn,E}$	2.0	ms	Decay time of excitatory synaptic current
$\tau_{syn,I}$	2.0	ms	Decay time of inhibitory synaptic current
V_{thresh}	-50.0	mV	Spike threshold
V_{reset}	-65.0	mV	Reset potential after a spike
τ_{ref}	0	ms	Duration of refractory period
Plasticity parameters			
A_+	10	pA	Maximum amplitude of weight increase
A_-	12	pA	Maximum amplitude of weight decrease
τ_+	20	ms	Time constant of the positive window
τ_-	20	ms	Time constant of the negative window
t_{exp}	0.25-10	s	Range of the STDP exposure times
Simulation and training parameters			
N	10-200		Range of reservoir sizes
Δt	0.1	ms	Simulator timestep
T	500	ms	Duration of signal slices and training steps
η	0.003	pA	Learning rate

makes it well-suited for neuromorphic platforms, where synaptic updates can be applied in parallel and in real time. Event-driven hardware implementations could exploit the inherent sparsity of spike-based communication, allowing for low-power, on-device inference in clinical settings. Future research should explore how variations in neuron and synapse parameters, as well as digital vs. analog substrates, affect the performance and energy consumption of STDP-enabled reservoir networks deployed on edge devices.

APPENDIX

The table III reports the parameters used for the LIF neuronal models (*IF_curr_exp* in pyNN), where only the membrane time constant vary from reservoir to readout neurons, STDP plasticity, simulation and training.

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